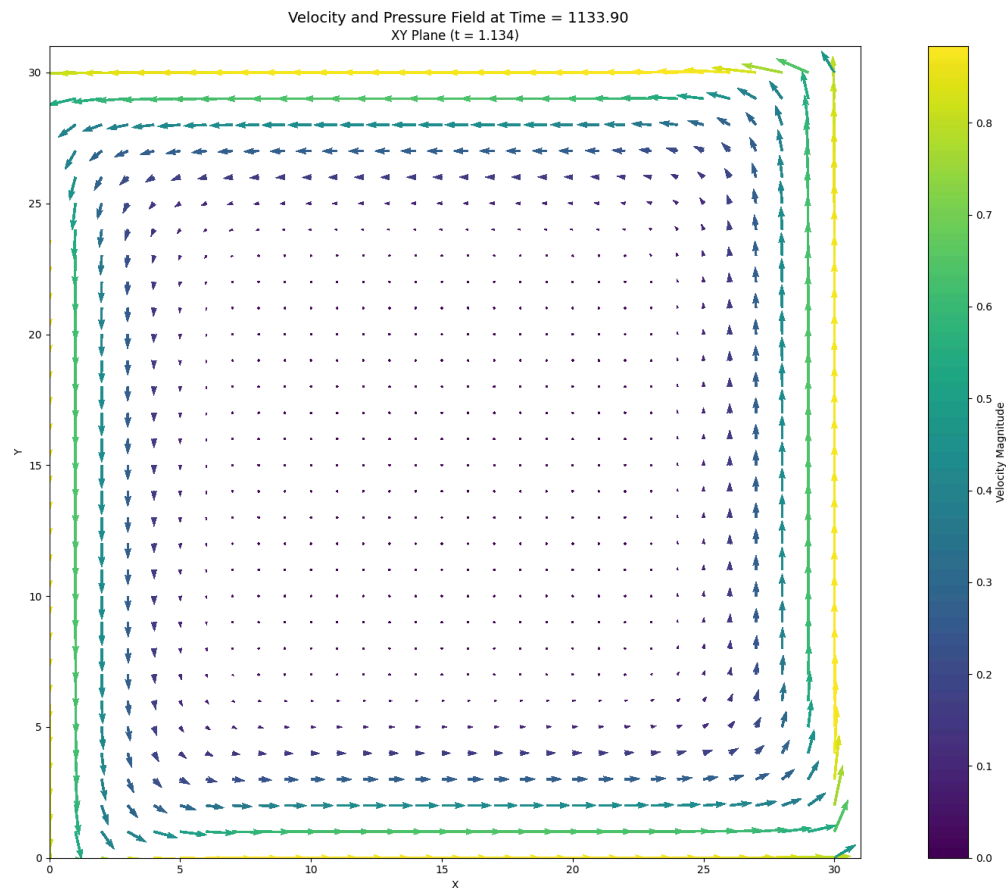


# Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



## 1. Simulation Parameters

| Quantity                | Symbol    | Value          | Unit                           |
|-------------------------|-----------|----------------|--------------------------------|
| Fluid density           | $\rho$    | 1.00000000e+03 | kg m <sup>-3</sup>             |
| Dynamic viscosity       | $\nu$     | 1.00000000e+01 | Pa s                           |
| Kinematic viscosity     | $\nu$     | 1.00000000e-02 | m <sup>2</sup> s <sup>-1</sup> |
| Characteristic velocity | $u_{max}$ | 1.00000000e+00 | m s <sup>-1</sup>              |
| Characteristic length   | $D$       | 1.00000000e+00 | m                              |
| Reynolds number         | $Re$      | 1.00000000e+02 | dimensionless                  |
| Body force              | $G$       | 9.81000000e+00 | m s <sup>-2</sup>              |
| Grid points in x        | $N_x$     | 3.10000000e+01 | cells                          |
| Grid points in y        | $N_y$     | 3.10000000e+01 | cells                          |
| Grid spacing in x       | $dx$      | 3.22580645e-02 | m                              |

| Quantity            | Symbol  | Value          | Unit |
|---------------------|---------|----------------|------|
| Grid spacing in y   | dy      | 3.22580645e-02 | m    |
| Time step           | dt      | 1.00000000e-04 | s    |
| Numerical threshold | epsilon | 7.00000000e-11 | -    |

## 2. Numerical Discretisation

| Quantity | Symbol | Value | Unit |
|----------|--------|-------|------|
|----------|--------|-------|------|

Parameters listed above are the values used by the selected solver.

## 3. Data and Pre-processing

| Quantity                             | Value          |
|--------------------------------------|----------------|
| Loaded field shape ( $t, x, y$ )     | (9851, 30, 31) |
| Regression field shape ( $t, x, y$ ) | (7451, 22, 27) |
| Regression samples                   | 4,425,894      |
| Dictionary terms                     | 5              |
| First saved time index               | 150            |
| Temporal crop at each boundary       | 1200 layers    |
| x-boundary crop                      | 4 points       |
| y-boundary crop                      | 2 points       |

## 4. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

Learned equation

$$u_t = -1.39217942e+00*uu_x - 9.75843704e-01*vu_y + 2.41619947e-02*u_{xx} + 1.02176557e-02*u_{yy} - 9.61020893e-04*p_x$$

| Term | Reference coefficient | Learned coefficient | Absolute error |
|------|-----------------------|---------------------|----------------|
| uu_x | -1.00000000e+00       | -1.39217942e+00     | 3.92179419e-01 |
| vu_y | -1.00000000e+00       | -9.75843704e-01     | 2.41562962e-02 |
| u_xx | +1.00000000e-02       | +2.41619947e-02     | 1.41619947e-02 |
| u_yy | +1.00000000e-02       | +1.02176557e-02     | 2.17655709e-04 |

| Term | Reference coefficient | Learned coefficient | Absolute error |
|------|-----------------------|---------------------|----------------|
| p_x  | -1.00000000e-03       | -9.61020893e-04     | 3.89791068e-05 |

## 5. STRidge Selection and Error Metrics

| Metric                              | Value          |
|-------------------------------------|----------------|
| Regularisation parameter, $\lambda$ | 1.00000000e-09 |
| Selected tolerance                  | 9.42668455e-04 |
| L0 penalty                          | 1.00000000e-06 |
| Training RMSE                       | 1.99802752e-02 |
| Validation RMSE                     | 2.52969478e-02 |

## 6. Reproducibility Notes

- Input directory: `not specified`
- Fields were loaded starting at saved time index `150`.
- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.
- The learned equation is compared term-by-term with the supplied reference coefficients.